Project ESCAPE: Does Entropy-Based Patching Find Syntax?

Final viva — Team Trimax

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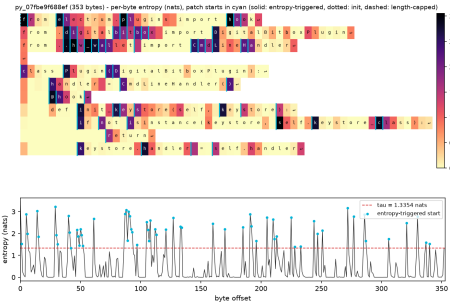
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The question

- BLT starts a new patch when its byte-level entropy model is uncertain: $H_i > \tau$.
- **Grammatical Branching Hypothesis:** patches start at AST / parse constituent starts more often than inside constituents — an average-case asymmetry, not a claim that starts are maximal-choice points.
- Tested on 16,300 files (Python, C++, English), held-out split 13,040, against
  - a **density-matched null**: each file's own patch lengths, shuffled, final patch fixed (10,000 permutations);
  - **structure-blind baselines**: newline, indentation change, word start.
- Five objectives (O1–O5), four predictions (P1–P4), plus an extension (a distilled patcher).

Stream A: a faithful patcher, then a distilled one

- HF port `itazap/blt-1b-hf` omits BLT's 512-byte sliding window; **we restored it** (8,192-token chunks), float32, $\tau = 1.3354$.
- Without it (HF full attention) the test margin collapses: $+0.023$ vs. $+0.120$ (Py), $+0.008$ vs. $+0.131$ (C++ <8 KB).
- 8 fidelity gates; logits match an independent reimplementation to 1.05×10^{-4}; amended gate set passed on Ada.
- Teacher outputs re-extracted on Ada; **per-file, score and diagnostic CSVs byte-identical** to the frozen run.
- Extension: 14.3M student distilled from the 99.5M patcher; golden gate reproduced within 0.0000 F1 on GPU.



Stream B: corpus, targets and controls

- 16,300 hash-verified files: 8,000 Python functions (CodeSearchNet), 8,000 C++ files (The Stack; CodeSearchNet has no C++), 300 WikiText-103 paragraphs; calibration 1,600/1,600/60, test 13,040.
- Targets: **tree-sitter** constituent starts (Python, C++), **benepar** parses (English); character offsets round-tripped to UTF-8 bytes; unique non-wrapper starts, outermost node wins.
- Structure-blind baselines from the same bytes: newline, indentation change, word start.
- O4 noise set: 20% of lines or keywords, 50% of identifiers; rebuilt at 5–40% for the rate sweep.
- O5 regions: tree-sitter queries for `*`, `&`, `new`, `delete`; audit reproduced 12,609/12,609 regions, an independent labeller agreed on 433/435 statements.
- Extension data: a disjoint 29k-file distillation corpus (pinned revision) and the pre-registered English holdout (4,107 paragraphs from 120 unused articles).

Ground truth from parsers: interactive tree views

- **Code:** tree-sitter gives an exact grammar tree; the highlighted nodes are the constituents we keep as targets (statement- and expression-level) with their start and end bytes.
- **English:** benepar gives a predicted phrase tree (NP, VP, PP); targets are phrase starts.
- Each viewer shows the full parse tree next to the source, as a self-contained HTML page (no server, no network).

Open a view (links work when the PDF is opened from the repository):

Python C++ English

Viewers: `stream_b/tree_viewer.py`, `stream_b/prose_tree_viewer.py`; examples in `stream_b/tree_views/`.

Stream C: frozen protocol

- One-to-one matching within k bytes; $k \in \{0, 1, 2, 4, 8\}$ chosen on the **calibration split only** $\rightarrow k = 0$ everywhere.
- Fixed-EOF null keeps patch-length multiset and boundary count; paired file bootstrap (2,000).
- Frozen teacher run scored once; later test-file runs (student, τ^*, $k = 1, 2$, O4 sweep, HF full attention, τ grid) **selected nothing**.
- New checks: repository-clustered bootstrap (Python, 315 repos) widens the CI 3.7$\times$ to [0.112, 0.127]; parser error recovery (28.0% of C++) changes the margin by $+0.004$ [-0.005, 0.015]; files $>$8,191 bytes by -0.002 [-0.008, 0.004].

P1: real alignment, but not beyond line/word segmentation

	F1	H_0 F1	Δ [95% CI]	F1/H_0	vs. best baseline
Python	0.152	0.032	+0.120 [0.117, 0.122]	4.7×	-0.242 (indentation)
C++	0.165	0.036	+0.129 [0.125, 0.134]	4.6×	-0.173 (word-start)
English	0.495	0.147	+0.348 [0.341, 0.354]	3.4×	-0.274 (word-start)

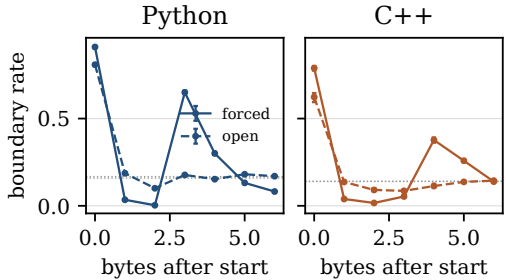
- P1 **supported** against the null ($p \leq 10^{-4}$ everywhere); against the best structure-blind baseline BLT underperforms it in all three languages.
- At BLT's density F1 is capped at $2/(r{+}1) = 0.184$ / 0.256 / 0.554; density-free check: word starts beat BLT on **both** precision and recall (Py 0.142/0.892 vs. 0.083/0.822).
- Precision $\ll$ recall (proposal predicted the reverse): every boundary has $\geq e^{\tau} \approx 3.8$ live continuations; identifier interiors are flat but not syntactic.
- P2 **supported**: end margins -0.017, -0.035, -0.079.

P3 reversed — prose constituents start at words

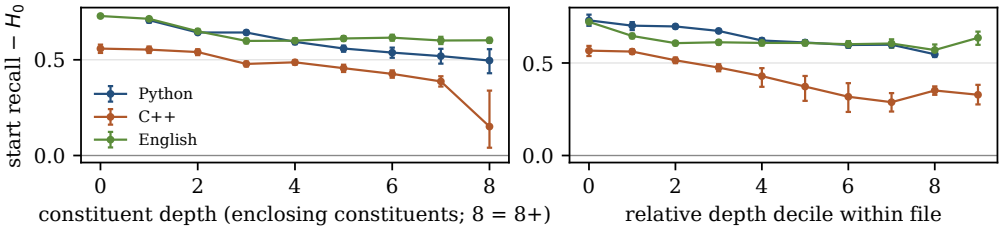
- Predicted: prose near chance. Observed: prose margin $+0.348 >$ code $+0.129$.
- Benepar constituents begin at words; the word-start rule alone gets 0.769 F1 in English.
- $\mathrm{F1}/H_0 = R/R_0$ is capped at $1/R_0 \approx$ mean patch length (5.8 / 7.1 / 3.8), so its $4.7\times$ / $4.6\times$ / $3.4\times$ favours code mechanically.
- Chance-corrected recall $(R - R_0)/(1 - R_0)$ ranks **English first**: 0.856 vs. 0.784 (Py), 0.588 (C++).
- BPP variance half of P3 holds: within-file variance 527.8 (C++), 69.4 (Py) vs. 10.3 (En).

P4 reversed at the start byte — confirmed at the forced bytes

- Per-type mean start margin: forced +0.759 vs. open +0.611 (Py) — opposite of P4.
- Test: boundary rate at bytes *after s*, same null draws (start rows reproduce the frozen recalls).
- Rest of keyword: -0.130 (Py), -0.088 (C++) vs. null.
- Byte after keyword: rate 0.0002 (Py), 0.035 (C++) vs. null 0.159 (Py) / 0.140 (C++); open-ended stays at the null.
- Boundary returns where choice reopens: Python condition start (rate 0.895); C++ after the forced ((rate 0.377).
- **Forcing is real; the start byte is just before it.**

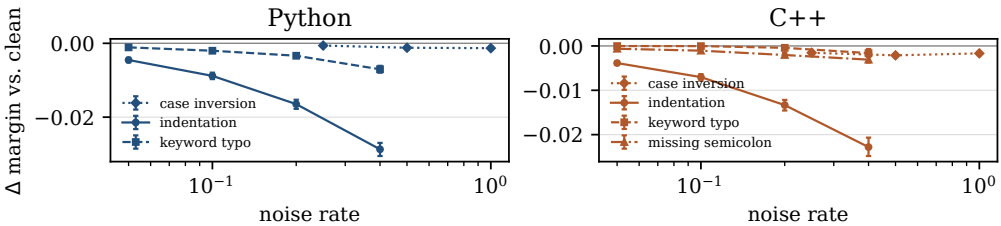


O3: depth on a common scale



- Raw tree-sitter depth counts language-specific scaffolding → count enclosing *constituents* instead, and normalise within file.
- Largest at the outermost constituents; 1 → 7 enclosing constituents: Py +0.707 → +0.519, C++ +0.553 → +0.387, En +0.714 → +0.601. Python > C++ at every depth.

O4: noise, now as a rate sweep



- Same 1,000 held-out files per language, each noise type at 5–40% (case 25–100%).
- Indentation does by far the most damage (Py -0.005 → -0.029, C++ -0.004 → -0.023); the others also grow with rate but stay small (typos -0.007 / -0.002, semicolons -0.003 at 40%). Typos cut BPP (-0.43 / -0.60). Every margin $\geq +0.086$.

O5: no evidence of risk-aware compute near memory-unsafe code

- Label audit: 12,609/12,609 regions reproduced; independent lexical labeller agrees on 433/435 statements.
- Frozen coefficient +0.0069 [0.0043, 0.0095] boundaries/byte — but:
- File FE only: **-0.0083 [-0.0127, -0.0039]** — unsafe statements get *fewer* boundaries (more predictable identifiers).
- Positive only after conditioning on identifier entropy; size 4.2% of the mean, 7% of a 1-SD identifier-entropy effect.
- Inverse BPP with file FE only: -0.0024 [-0.0066, 0.0017] (null).
- Reading: a small conditional effect, not risk awareness.

Beyond the proposal: a 7× smaller distilled patcher

	Teacher Δ	Student Δ	kept	Student−teacher [95% CI]
Python	+0.120	+0.109	91%	-0.011 [-0.011, -0.010]
C++	+0.129	+0.119	92%	-0.011 [-0.012, -0.009]
English	+0.348	+0.304	87%	-0.044 [-0.048, -0.040]

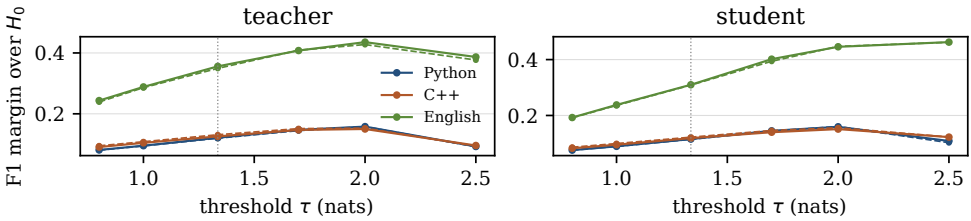
- Online KD ($\alpha = 0.1$, $T = 2$) on a disjoint 29k-file corpus; val boundary F1 vs. teacher 0.871; no-teacher control ($\alpha = 1$) 0.789.
- Full 16,300-file extraction + frozen C-v1 scoring with its own calibration ($k = 0$).
- Native teacher minus teacher on 512-byte windows: +0.004 (Py) / +0.004 (C++) F1, 38% / 43% of the student's deficit; capacity explains the rest.
- Cost: forward 5.0× faster than the teacher on a RTX 2080 Ti.

Follow-ups: byte-balanced student, adapted teacher

	Balanced−student [95% CI]	Adapted−student [95% CI]	bar
Python	-0.002 [-0.003, -0.001]	+0.006 [0.004, 0.008]	0.017
C++	-0.002 [-0.0035, -0.0001]	+0.005 [0.003, 0.007]	0.022
English	+0.010 [0.008, 0.012]	+0.029 [0.026, 0.032]	0.013

- Paired Δ differences, $k = 0$, on the pre-registered 600-file code subset and the 240 English test files (exploratory: full-corpus extraction did not fit).
- Byte-balanced sampling trades code for prose: English up, code slightly down; window F1 vs. teacher 0.860 (distilled 0.871).
- Adapted teacher (99.5M, same loss and data): window F1 0.955; pre-registered verdict **not met** (1 of 3 domains beat the student's CI width).
- In code it matches the unadapted teacher read through the same windows: it recovers the teacher rather than improving on it.

Threshold sweep and the confirmatory English holdout



- Pre-registered τ^* on calibration: teacher 2; student 2/2/2.5.
- One test run at τ^*: teacher Δ +0.153 (Py), +0.158 (C++), +0.426 (En) — still below baselines (-0.210).
- Student at its own τ^*: Δ +0.147, +0.156, +0.460 — also below baselines.
- Later, exploratory: grid on test (dashed) peaks at the same τ^* (3/3, 3/3); every τ trails the best baseline by $\geq$0.148 F1.
- English holdout (4,107 unseen paragraphs, scored once): teacher +0.350, student +0.305; vs. word starts -0.273.

Verdict on H1 and the proposal's deliverables

- **Criteria from the proposal:** P1 beats both baselines, P2 holds, P3 dissociates code from prose, P4 stratification appears.
- **Outcome:** H1 is **not supported** under this criterion: only P2 holds as stated.
  - P1: beats the density-matched null everywhere, but not the whitespace and word-start baselines.
  - P2: holds (end margins negative).
  - P3: reversed, also under chance-corrected recall.
  - P4: reversed at the start byte; the mechanism holds at the bytes after the keyword.
  - Precision $>$ recall (proposal's mechanism): fails.
- **Deliverables:** (1) extraction pipeline `stream_a`; (2) tree-sitter / benepar mapping with memory-unsafe queries `stream_b`; (3) harness with P1–P4, both baselines, permutation test, depth and type stratification `escape_scoring`, `escape_eval`; (4) entropy heatmaps and BPP profiles; (5) report and these slides.
- **Released models:** student, no-teacher control (α=1), byte-balanced student on Hugging Face; checkpoints and curves in the W&B project.

Limitations and future work

- **Limitations:** third-party checkpoint with a restored window (no bitwise check against the gated original); English targets are predicted parses; O5 labels are syntactic; one student size, one seed; follow-up arms scored on a subset; no subword-tokenizer baseline for O4; no multiple-comparison correction; English intervals resample paragraphs, not articles.
- **Future work:** parser-supervised teacher; subword baseline for O4; several student seeds; full-corpus scoring of the follow-up models; single-pass student kernel; bitwise comparison with the original checkpoint.

Conclusions

- Entropy boundaries track constituent starts **3.4–4.7× above chance** — held-out and on the English holdout; a 7× smaller student keeps ≈90% of the margin.
- They do **not** beat indentation / word-start segmentation at any τ or k.
- Each failed prediction has a mechanical explanation:
  - precision $<$ recall: several boundaries per target;
  - P3: prose constituents begin at words, where the patcher fires;
  - P4: forcing acts after the opener, not at its start;
  - O5: the "residual" appears only after an identifier-entropy control.
- Entropy is a genuine but shallow signal of word and line onsets — syntax shapes it without being what it measures.